COURSE PLANNING (RPKPS)
UNIVERSITAS MULTIMEDIA NUSANTARA

VALIDATION PAGE

Course Name : Advanced Web Programming
Course Code : IF451
Course Coordinator : Wirawan Istiono, S.Kom, M.Kom
Team of Lecturers :

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2	PUTU ANANTA YOGISWARA, S.KOM, M.T.I.	L00698	
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On behalf of the team,
Date: 14 August 2025

(Wirawan Istiono, S.Kom, M.Kom)
Course Coordinator

Approved by
Date:

(Arya Wicaksana, S.Kom., M.Eng.Sc)
Head of Department/Program

Has been checked and considered comply to UMN standard
Date:

(David Agustriawan S.Kom., M.Sc., Ph.D)
Internal Quality Assurance Office

INFORMATICS
FACULTY OF ENGINEERING AND INFORMATICS
ACADEMIC YEAR 2025/2026

COURSE PLANNING (RPKPS)
UNIVERSITAS MULTIMEDIA NUSANTARA

COURSE NAME	: Advanced Web Programming
CODE / CREDIT	: IF451 / 3
SEMESTER	: 5
PREREQUISITE	: IF352 – Basic Web Programming
COURSE STATUS	: Mandatory/~~elective~~

A. COURSE DESCRIPTION

In the Web Programming course, students will learn concepts and techniques for programming web applications on the server side using the Node programming language connected to the MySQL database. The topics and material presented include basic NodeJS syntax, NodeJS control structures, library functions, object-oriented programming in Node JS, the Node JS Framework, and API development. The Web Programming course gives students the competency to develop server-side applications connected to databases..

B. LEARNING OUTCOME

B.1. Program Expected Learning Outcomes (ELO) Related to the Course

IQF Level: 6

ELO 3 Professional Skills (Entrepreneurial spirit in technology domain): Students are expected to demonstrate initiative, creativity, and business acumen to identify opportunities and drive innovation within the technology sector.

ELO 7 Software Design and Implementation Abilities: Students are expected to design, develop, and maintain software solutions following best engineering practices and industry standards.

B.2. Course Learning Outcomes (CLO)

After passing this course, students will be able to use the skills and knowledge from this course as a Web Developer in intermediate level with competences as follow:

ELO 3	CLO031	Students are able to apply informatics science in implementing using programming languages in Advanced Web Programming
ELO 7	CLO071	Students are able to combine technical skills and entrepreneurial concepts in meeting business demands and societal demands.

B.3. Course Sub Learning Outcomes (Sub-CLO)

CLO071	Sub-CLO0711	Students are able to explain the basic concepts of web development. (C2)
CLO071	Sub-CLO0712	Students are able to build basic web programming and feature in Node JS (C6)
CLO071	Sub-CLO0713	Students are able to implement various types of libraries and database to create web application using Node JS (C3)
CLO071	Sub-CLO0714	Students are able to implement Node JS Framework to build Web application (C3, C6)
CLO031	Sub-CLO0311	Students are able to collaborate and demonstrate the results of web project that has been created through the online domain. (C6)

C. LEARNING ANALYSIS

-Figure is attached-

D. TOPICS

1. Introduction to Node.js & Environment Setup
2. Modules & Packages in Node.js
3. JSON Basics & CRUD with Files
4. Mini Project: Student Management (JSON)
5. Introduction to MySQL Database
6. Connecting Node.js to MySQL Database
7. CRUD Operations in MySQL Database
8. Introduction to Express.js
9. Middleware & Routing in Express
10. REST API with Express + MySQL Database
11. Connecting Express API to ReactJS
12. File Upload & Basic Authentication
13. Fullstack Mini Project Case Study
14. Final Project Presentation

E. EVALUATION

1. Attending lectures punctually is mandatory. Students will be considered absent if coming over the specified time.
2. Attending 14 lectures is mandatory. Attending a minimum of 11 from 14 meetings is required to be able to take the final test.

3. Final grade is determined by following components:

   a. Midterm Test : 30%
   b. Final Test : 40%
   c. Quiz, & presenting : 30%

Table: Assessment Distribution of Learning Outcomes

ELO	CLO	Sub-CLO	Quiz & Presenting & Lab Assignment		Midterm Ex.		Final Ex.	
			Lecture	Lab	Lecture	Lab	Lecture	Lab
ELO7	**CLO071**	**Sub-CLO0711**	**2.5%**	**2.5%**	**5%**	**5%**	**2.5%**	**2.5%**
ELO7	**CLO071**	**Sub-CLO0712**	**2.5%**	**2.5%**	**5%**	**5%**	**5%**	**5%**
ELO7	**CLO071**	**Sub-CLO0713**	**2.5%**	**2.5%**	**5%**	**5%**	**5%**	**5%**
ELO7	**CLO071**	**Sub-CLO0714**	**2.5%**	**2.5%**			**5%**	**5%**
ELO3	**CLO031**	**Sub-CLO0311**	**5%**	**5%**			**2.5%**	**2.5%**
Total			**15%**	**15%**	**15%**	**15%**	**20%**	**20%**

FINAL GRADING:

Score	Alphabetical Grade	Numerical Grade	Remarks
85 – 100	A	4	Excellent
80 – 84,99	A-	3,7	Good
75 – 79,99	B+	3,3	
70 – 74,99	B	3,0	
65 – 69,99	B-	2,7	Satisfactory
60 – 64,99	C+	2,3	
55 – 59,99	C	2,0	
45 – 54,99	D	1,0	Poor
0 – 44,99	E	0	Very Poor
	F	0	Academic Violation

F. REFERENCE AND RESOURCES

-Main-

1. B. Lawson and J. Encinas, Node.js Web Development, 6th ed., Packt Publishing, 2023
2. R. Raj, Learning Node.js Development, Packt Publishing, 2018.
3. E. Cantelon, M. Harter, T. Holowaychuk, and N. Rajlich, Node.js in Action, 2nd ed., Manning Publications, 2017.
4. Maulana, a., bau, r.t.r., munawar, z., setiawan, r., aisa, s., istiono, w., irfan, a., pratama, y.a., ramadhany, e.d., pomalingo, s. And permana, a.a., 2023. *Pemrograman web 101: memahami dasar-dasar untuk mengembangkan situs web*. Get press indonesia.

-Additional/Supporting-

5. The Node.js Foundation, “Node.js Documentation,” [Online]. Available: https://nodejs.org/en/docs/. [Accessed: Jun. 5, 2025].
6. Express.js Team, “Express.js Documentation,” [Online]. Available: https://expressjs.com/. [Accessed: Jun. 5, 2025].

H. WEEKLY LESSON PLAN

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	*Assessment*			Ref.
					Assessment type and Grading System	Indicators	Weight	
1.	Sub-CLO0711 Students are able to explain the basic concepts of web development. (C2)	<u>Topics</u>: Introduction to Node.js & Environment Setup <u>Sub-topics</u>: - *Backend vs frontend.* - *Role of Node.js in web development.* - *Installing Node.js, npm, and VSCode.* - *Running JavaScript in Node.js.*	**<u>Learning Methods</u>: Lectures** **<u>Activities</u>:** Pre-class : - Material Reading In class: (Theory & Lab) - Material Presentation and Discussion - Practical on the computer Post-class: (Theory & Lab) - Writing Code - Forum Discussion	 2x50' 2x60' 2x60' 170'			Mid 2.5% Final: 2.5%	Ref1 – ch1 Ref2 – ch1 Ref4 – ch1 Ref5 - online
2.	Sub-CLO0712 Students are able to build basic web programming and feature in Node JS (C6)	<u>Topics</u>: Modules & Packages in Node.js <u>Sub-topics</u>: - *Basic concept NodeJS* - *Built-in modules (fs, path, os).* - *Creating custom modules (module.exports).* - *Installing packages with npm*	**<u>Learning Methods</u>: Lectures** **<u>Activities</u>:** Pre-class : - Material Reading In class: (Theory & Lab) - Material Presentation and Discussion - Practical on the computer Post-class: (Theory & Lab) - Writing Code - Forum Discussion	 2x50' 2x60' 2x60' 170'	**<u>Lecture Assessment</u>:** Quiz 1 **<u>Grading system</u>**: Points per question **<u>Lab Assessment</u>:** Structured Tasks **<u>Grading system</u>**: Manual Task Solution	Accuracy in answering questions The number of resolved application project problems and the suitability of problem solving to the problem	Quiz1 5% Lab Assessment 2.5% Mid 2.5% Final: 2.5%	Ref1 – Ch6 Ref2 – ch2 Ref5 - online
3.	Sub-CLO0712 Students are able to build basic web programming and feature in Node JS (C6)	<u>Topics</u>: JSON Basics & CRUD with Files <u>Sub-topics</u>: - *JSON structure.* - *Reading & writing JSON files.* - *Adding, updating, deleting entries.*	**<u>Learning Methods</u>: Lectures** **<u>Activities</u>:** Pre-class : - Material Reading In class: (Theory & Lab) - Material Presentation and Discussion - Practical on the computer	 2x50' 2x60'	**<u>Lab Assessment</u>:** Structured Tasks **<u>Grading system</u>**: Manual Task Solution	The number of resolved application project problems and the suitability of problem solving to the problem	Lab Assessment 2.5% Mid 5% Final: 5%	Ref1 - ch3 Ref2 – ch3 Ref5 - online

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	*Assessment*			Ref.
					Assessment type and Grading System	Indicators	Weight	
			Post-class: (Theory & Lab) - Writing Code - Forum Discussion	2x60' 170'				
4.	Sub-CLO0713 Students are able to implement various types of libraries and database to create web application using Node JS (C3)	Topics: Mini Project: Student Management (JSON) Sub-topics: - *Create a Node.js CLI or server app to manage student data.* - *Implement add, list, update, delete features.*	**Learning Methods: Lectures** **Activities:** Pre-class : - Material Reading In class: (Theory & Lab) - Material Presentation and Discussion - Practical on the computer Post-class: (Theory & Lab) - Writing Code - Forum Discussion	 2x50' 2x60' 2x60' 170'	**Lecture Assessment:** Quiz2 **Grading system**: Points per question **Lab Assessment:** Structured Tasks **Grading system**: Manual Task Solution	Accuracy in answering questions The number of resolved application project problems and the suitability of problem solving to the problem	Quiz2 5% Lab Assessment 5% Mid 5% Final: 5%	Ref1 – ch4 Ref3 – ch3 Ref5 - online
5.	Sub-CLO0713 Students are able to implement various types of libraries and database to create web application using Node JS (C3)	Topics: Introduction to MySQL Database Sub-topics: - *SQL vs NoSQL.* - *Database – Collection - Document.* - *MySQL Database Atlas account setup (cloud database)*	**Learning Methods: Lectures** **Activities:** Pre-class : - Material Reading In class: (Theory & Lab) - Material Presentation and Discussion - Practical on the computer Post-class: (Theory & Lab) - Writing Code - Forum Discussion	 2x50' 2x60' 2x60' 170'			Mid 5% Final: 2.5%	Ref1 – ch5 Ref3 – ch4 Ref4 – ch2 Ref5 - online
6.	Sub-CLO0713 Students are able to implement various types of libraries and database to create web application using Node JS (C3)	Topics: Connecting Node.js to MySQL Database Sub-topics: - *MySQL Database Node.js driver & Mongoose ODM.* - *Inserting data into MySQL Database.*	**Learning Methods: Lectures** **Activities:** Pre-class : - Material Reading In class: (Theory & Lab) - Material Presentation and Discussion - Practical on the computer	 2x50' 2x60'	**Lab Assessment:** Structured Tasks **Grading system**: Manual Task Solution	The number of resolved application project problems and the suitability of problem solving to the problem	Lab Assessment 2.5% Mid 5% Final: 2.5%	Ref1 – ch6 Ref2 – ch6 Ref5 - online

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	*Assessment*			Ref.
					Assessment type and Grading System	Indicators	Weight	
			Post-class: (Theory & Lab) - Writing Code - Forum Discussion	2x60' 170'				
7.	Sub-CLO0713 Students are able to implement various types of libraries and database to create web application using Node JS (C3)	<u>Topics</u>: CRUD Operations in MySQL Database <u>Sub-topics</u>: - *Find, findOne, insertOne, updateOne, deleteOne*	**<u>Learning Methods</u>: Lectures** **<u>Activities</u>:** Pre-class : - Material Reading In class: (Theory & Lab) - Material Presentation and Discussion - Practical on the computer Post-class: (Theory & Lab) - Writing Code - Forum Discussion	 2x50' 2x60' 2x60' 170'	**<u>Lab Assessment</u>:** Structured Tasks **<u>Grading system</u>**: Manual Task Solution	The number of resolved application project problems and the suitability of problem solving to the problem	Lab Assign 2.5% Mid 5% Final: 2.5%	Ref1 – ch7 Ref2 – ch7 Ref5 - online
<u>Midterm Test</u> Type or Assignment Methods : Project							30%	Ref
8.	Sub-CLO0714 Students are able to implement Node JS Framework to build Web application (C3, C6)	<u>Topics</u>: Introduction to Express.js <u>Sub-topics</u>: - *Installing Express.* - *Route handling (GET, POST).* - *Sending JSON responses*	**<u>Learning Methods</u>: Lectures** **<u>Activities</u>:** Pre-class : - Material Reading In class: (Theory & Lab) - Material Presentation and Discussion - Practical on the computer Post-class: (Theory & Lab) - Writing Code - Forum Discussion	 2x50' 2x60' 2x60' 170'	**<u>Lecture Assessment</u>:** Quiz3 **<u>Grading system</u>**: Points per question **<u>Lab Assessment</u>:** Structured Tasks **<u>Grading system</u>**: Manual Task Solution	Accuracy in answering questions The number of resolved application project problems and the suitability of problem solving to the problem	Quiz3 5% Lab Assessment 2% Final: 2.5%	Ref1-ch8 Ref2 – ch8 Ref4 – ch4 Ref6 - online
9.	Sub-CLO0714 Students are able to implement Node JS Framework to build Web application (C3, C6)	<u>Topics</u>: Middleware & Routing in Express <u>Sub-topics</u>:	**<u>Learning Methods</u>: Lectures** **<u>Activities</u>:** Pre-class :		**<u>Lab Assessment</u>:** Structured Tasks **<u>Grading system</u>**:	The number of resolved application project problems and the	Lab Assessment 2% Final:	Ref1 – ch9

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assessment type and Grading System	Indicators	Weight	Ref.
		- *Static files serving.* - *Built-in & custom middleware*	- Material Reading In class: (Theory & Lab) - Material Presentation and Discussion - Practical on the computer Post-class: (Theory & Lab) - Writing Code - Forum Discussion	 2x50' 2x60' 2x60' 170'	Manual Task Solution	suitability of problem solving to the problem	2%	Ref3 – ch5 Ref6 - online
10.	Sub-CLO0714 Students are able to implement Node JS Framework to build Web application (C3, C6)	Topics: REST API with Express + MySQL Database Sub-topics: - *GET, POST, PUT, DELETE routes.* - *Data validation.*	**Learning Methods: Lectures** **Activities:** Pre-class : - Material Reading In class: (Theory & Lab) - Material Presentation and Discussion - Practical on the computer Post-class: (Theory & Lab) - Writing Code - Forum Discussion	 2x50' 2x60' 2x60' 170'	**Lab Assessment:** Structured Tasks **Grading system:** Manual Task Solution	The number of resolved application project problems and the suitability of problem solving to the problem	Lab Assessment 2% Final: 2%	Ref1 – ch10 Ref2 – ch10 Ref4 – ch5 Ref6 - online
11.	Sub-CLO0713 Students are able to implement various types of libraries and database to create web application using Node JS (C3) Sub-CLO0714 Students are able to implement Node JS Framework to build Web application (C3, C6)	Topics: Connecting Express API to ReactJS Sub-topics: - *CORS handling.* - *Fetch API in ReactJS*	**Learning Methods: Lectures** **Activities:** Pre-class : - Material Reading In class: (Theory & Lab) - Material Presentation and Discussion - Practical on the computer Post-class: (Theory & Lab) - Writing Code - Forum Discussion	 2x50' 2x60' 2x60' 170'	**Lab Assessment:** Structured Tasks **Grading system:** Manual Task Solution	The number of resolved application project problems and the suitability of problem solving to the problem	Lab Assessment 2% Final: 2%	Ref1 – ch11 Ref3 – ch5 Ref6 - online
12.	Sub-CLO0713 Students are able to implement various types of libraries and	Topics: File Upload & Basic Authentication	**Learning Methods: Lectures** **Activities:**		**Lecture Assessment:** Quiz 1	Accuracy in answering questions	Quiz4 5% Final:	Ref1 – ch12

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	*Assessment* Assessment type and Grading System	Indicators	Weight	Ref.
	database to create web application using Node JS (C3) Sub-CLO0714 Students are able to implement Node JS Framework to build Web application (C3, C6)	Sub-topics: - *File upload using multer.* - *Login with username/password stored in MySQL Database*	Pre-class : - Material Reading In class: (Theory & Lab) - Material Presentation and Discussion - Practical on the computer Post-class: (Theory & Lab) - Writing Code - Forum Discussion	2x50' 2x60' 2x60' 170'	**Grading system**: Points per question		2%	Ref3 – ch6 Ref4 – ch7 Ref6 - online
13.	Sub-CLO0712 Students are able to build basic web programming and feature in Node JS (C6) Sub-CLO0714 Students are able to implement Node JS Framework to build Web application (C3, C6)	Topics: Fullstack Mini Project Case Study Sub-topics: - *Backend: CRUD API + file upload + login.* - *Frontend: ReactJS interface to manage students*	**Learning Methods: Lectures** **Activities:** Pre-class : - Material Reading In class: (Theory & Lab) - Material Presentation and Discussion - Practical on the computer Post-class: (Theory & Lab) - Writing Code - Forum Discussion	2x50' 2x60' 2x60' 170'	**Lab Assessment:** Structured Tasks **Grading system**: Manual Task Solution	The number of resolved application project problems and the suitability of problem solving to the problem	Lab Assessment 2% Final: 2%	Ref1. Ref2 Ref3 – ch7 Ref6 - online
14.	*Sub-CLO0311* Students are able to show and demonstrate the results of web project that has been created through the online domain. (C3)	Topics: Final Project Presentation Sub-topics: - Class Presentation project	**Learning Methods: Lectures** **Activities:** Pre-class : - Preparing presentation In class: (Theory & Lab) - Project Presentation Post-class: (Theory & Lab) - Forum Discussion	2x50' 2x60' 2x60' 170'	**Lecturers Assessment:** Presentation **Grading system**: Grading Rubric **Lab Assessment (Presentation):** Presentation **Grading system**: Grading Rubric	the process of working on the web and presentation and communication skills to explain the web project that has been created	Presentation: 10% Lab Assess (Presentation): 5% Final: 5%	

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	*Assessment*			Ref.
					Assessment type and Grading System	Indicators	Weight	
<u>Final Test</u> Type or Assignment Methods: Website Project							40%	Ref

I. Task/Project Details:

1. Quiz and Presentation no: 1, 2, 3, 4, Week 2, 4, 8, 12

Course :	Advanced Web Programming (Lecture)	Course Code :	IF451
Task/Project Name :	Quiz 1, 2, 3, 4 and Presentation	Weight :	30% (Quiz: 20%, Presentation 10%)
Related Sub-CLO :	Sub-CLO 0711, 0712, 0713, 0714		
A. Individual Activities			
Description	Students read lecture materials and read reference sources and tutorials provided to answer question in quiz		
B. Structured Work/Task			
Assessment Type :	Quiz		
Description :	Students take the Quiz in accordance with the related Sub-CLO		
Output and format :	**<u>Output</u>:** Multiple-choice, true-false, essay or program (coding) **<u>Format:</u>** Quiz answers on e-learning		
Indicators, Criteria, and Weight :	a. **1st Indicator (Weight 100%)** Answers according to the choice of system (multiple-choice / true-false), accuracy of answers (essay) or suitability to the question (program)		
Project/Work Timeline :	**Total duration:** With details: a. Sub-CLO 0711 b. Sub-CLO 0712 c. Sub-CLO 0713 d. Sub-CLO 0714 e. Sub-CLO 0311	**10-30 minutes, before or after lectures** Week 2 Week 4 Week 8 Week 12 Week 14 (Presentation)	
Others :	-		
References :	-		

2. Lab Assessment no: 2, 3, 4, 6, 7, 10, 12 Week 2, 4, 8, 12

Course	:	Advanced Web Programming (Laboratory)	Course Code		IF451
Task/Project Name	:	Lab Assessment 2,3,4,6,7,10,12	Weight	:	30 %
Related Sub-CLO	:	Sub-CLO 0711, 0712, 0713, 0714			

C. Individual Activities			
Description		Student create web projects and complete all tasks that have been given in the material	
D. Structured Work/Task			
Assessment Type	:	Task and challenge to solve web project problem	
Description	:	Student groups are given the task of making web or modification web according the question that have been given in the assessment.	
Output and format	:	**<u>Output</u>:** Script/Program **<u>Format:</u>** web project or script program	
Indicators, Criteria, and Weight	:	**a. 1st Indicator (Weight 100%)** Student complete all task and challenge to solve web project problem	
Project/Work Timeline	:	**Total duration:** With details: Sub-CLO0711 Sub-CLO0712 Sub-CLO0713 Sub-CLO0714 Sub-CLO0311	**1 week** Week 2, 3 Week 6,7 Week 4 Week 8, 9, 10, 11, 13 Week 14 (Presentation)
Others	:	-	
References	:	-	

3. Mid Exam

Course	:	Advanced Web Programming (Lecture and Laboratory)	Course Code		IF451
Task/Project Name	:	Mid exam	Weight	:	30%,
Related Sub-CLO	:	Sub-CLO 0711, 0712, 0713			

	E. Individual Activities
Description	Student create website projects and complete all tasks that have been given in the material
	F. Structured Work/Task
Assessment Type :	Task and challenge to solve website project problem
Description :	Student groups create website project using NodeJS according the project plan that they give in the class
Output and format :	**<u>Output</u>:** Script/Program **<u>Format:</u>** Website project or script program
Indicators, Criteria, and Weight :	**b. 1st Indicator (Weight 100%)** Student create a website according to the rules and provisions of the question
Project/Work Timeline :	**Total duration:** **1 week** With details: Sub-CLO0711 — Mid exam Sub-CLO0712 — Mid exam Sub-CLO0713 — Mid exam
Others :	-
References :	-

4. Final Exam

Course	:	Advanced Web Programming (Lecture and Laboratory)	Course Code	IF451
Task/Project Name	:	Mid exam and Final exam	Weight :	40%,
Related Sub-CLO	:	Sub-CLO 0711, 0712, 0713, Sub-CLO0311		

G. Individual Activities

Description		Task and challenge to solve website project problem

H. Structured Work/Task

Assessment Type	:	Task and challenge to solve website project problem	
Description	:	Student groups continue website project using NodeJS framework according the project plan that they give in the class	
Output and format	:	**Output:** Script/Program **Format:** Website project or script program	
Indicators, Criteria, and Weight	:	**c. 1st Indicator (Weight 100%)** Student create a website according to the rules and provisions of the question	
Project/Work Timeline	:	**Total duration:** With details: Sub-CLO0711 Sub-CLO0712 Sub-CLO0713 Sub-CLO0714 Sub-CLO0311	**1 week** Final exam Final exam Final exam Final exam Final exam
Others	:	-	
References	:	-	

Attachment: Learning Outcome Analysis

Learning Outcome Stages Chart

Program Expected Learning Outcomes

Final Test/Evaluation

Sub-CLO0311 - Students are able to collaborate and demonstrate the results of web project that has been created through the online domain. (C6)

Sub-CLO0714 - Students are able to implement Node JS Framework to build Web application (C3, C6)

Midterm Test/Evaluation

Sub-CLO0713 - Students are able to implement various types of libraries and database to create web application using Node JS (C3)

Sub-CLO0712 - Students are able to build basic web programming and feature in Node JS (C6)

Sub-CLO0711 - Students are able to explain the basic concepts of web development. (C2)

Prerequisite Course: IF352 – Basic Web Programming

J. Revision History

Course Code	Revision No	Date in Effect	Changes
IF451	1	2025/06/05	Customize ELO, CLO, Sub-CLO and new material
IF451	2	2025/14/08	Customize material and sub-CLO